

Dr. Rohan Mukherjee

Assistant Professor of Computational Neuroscience
| Spiking Networks & Learning

CONTACT

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AFFILIATIONS

Society for Neuroscience

Bernstein Network

SKILLS & METHODS

- Computational modelling
- Bayesian inference
- Dynamical systems
- Large-scale simulation
- Python JAX
- PyTorch Julia
- HPC / SLURM

LANGUAGES

English · Bengali · Hindi

MEMBERSHIPS

ACM

Cosyne

Dr. Rohan Mukherjee

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Research Statement

My research develops biologically plausible models of learning in spiking neural networks, bridging theoretical neuroscience and machine learning to understand how the brain achieves sample-efficient generalisation.

Education

Ph.D., Computational Neuroscience 2019
Indian Institute of Science (IISc)

M.Sc., Physics 2013
IIT Bombay

Appointments

Assistant Professor 2021 – Present
IIT Delhi — Dept. of Computer Science

- Built a research group of 5 PhD students and 2 postdocs working on biologically plausible learning.
- Raised ₹1.2 crore in competitive funding and published 3 first-author papers in the first three years.

Postdoctoral Fellow 2019 – 2021
University College London — Gatsby Unit

- Developed local plasticity rules that approximate backpropagation, published at NeurIPS 2023.

Selected Publications

- [1] Mukherjee R., Chen L., Rao S. “Sample-efficient credit assignment in spiking networks.” Nature Machine Intelligence 2024
- [2] Mukherjee R., Patel A. “Local plasticity rules approximate backpropagation.” NeurIPS 2023
- [3] Mukherjee R., et al. “A dynamical systems view of working memory.” eLife 2022

Grants, Awards & Honors

- SERB Core Research Grant (₹1.2 Cr, 2022–2025)
- Wellcome Trust Early-Career Fellowship (2019)

Teaching

CS6480: Neural Computation (graduate) 2021–present
IIT Delhi

CS2100: Machine Learning Foundations 2022–present
IIT Delhi

Presentations

- Invited talk: Plausible deep learning — Cosyne 2024
- Symposium: Neuromorphic RL — NeurIPS Workshop 2023